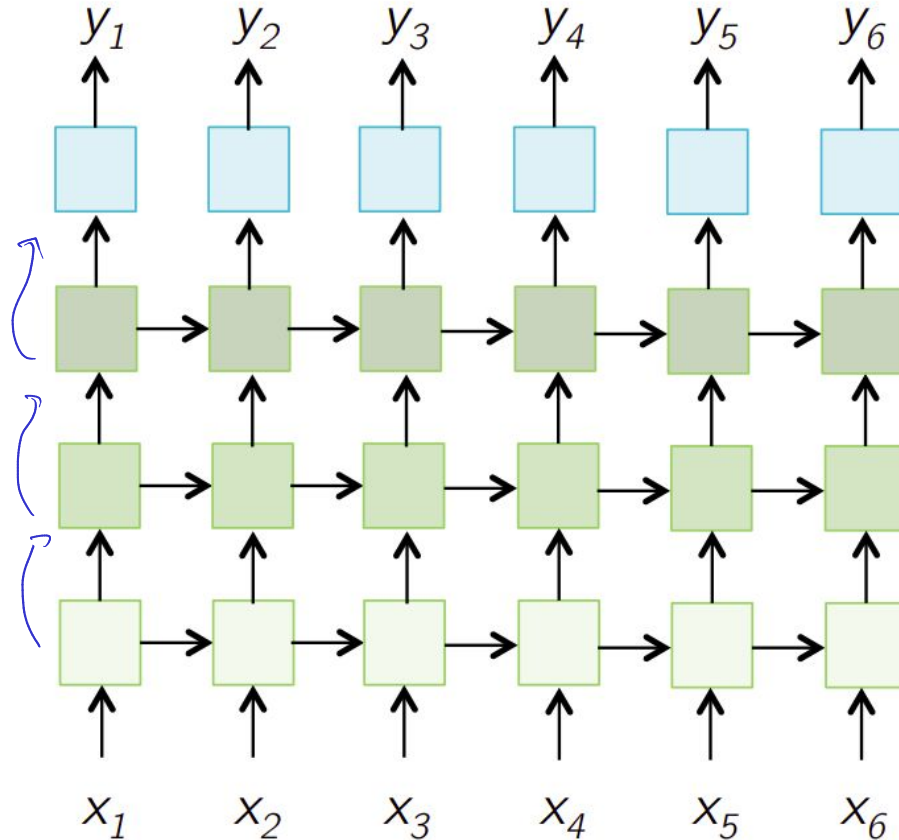
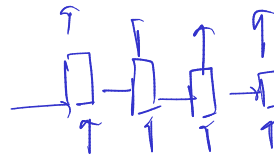
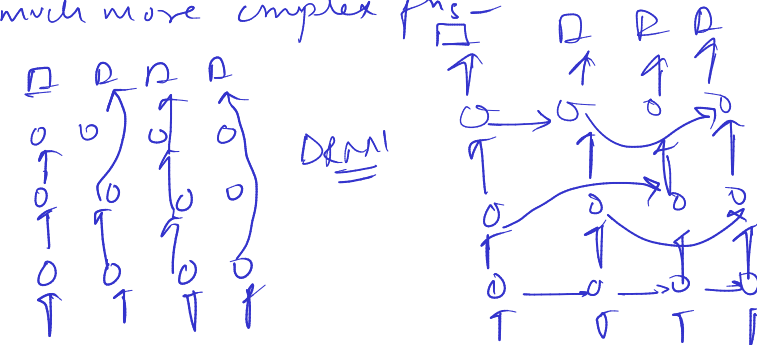




make deep (stacked) recurrent neural networks more interesting:

improves the expressivity of RNN

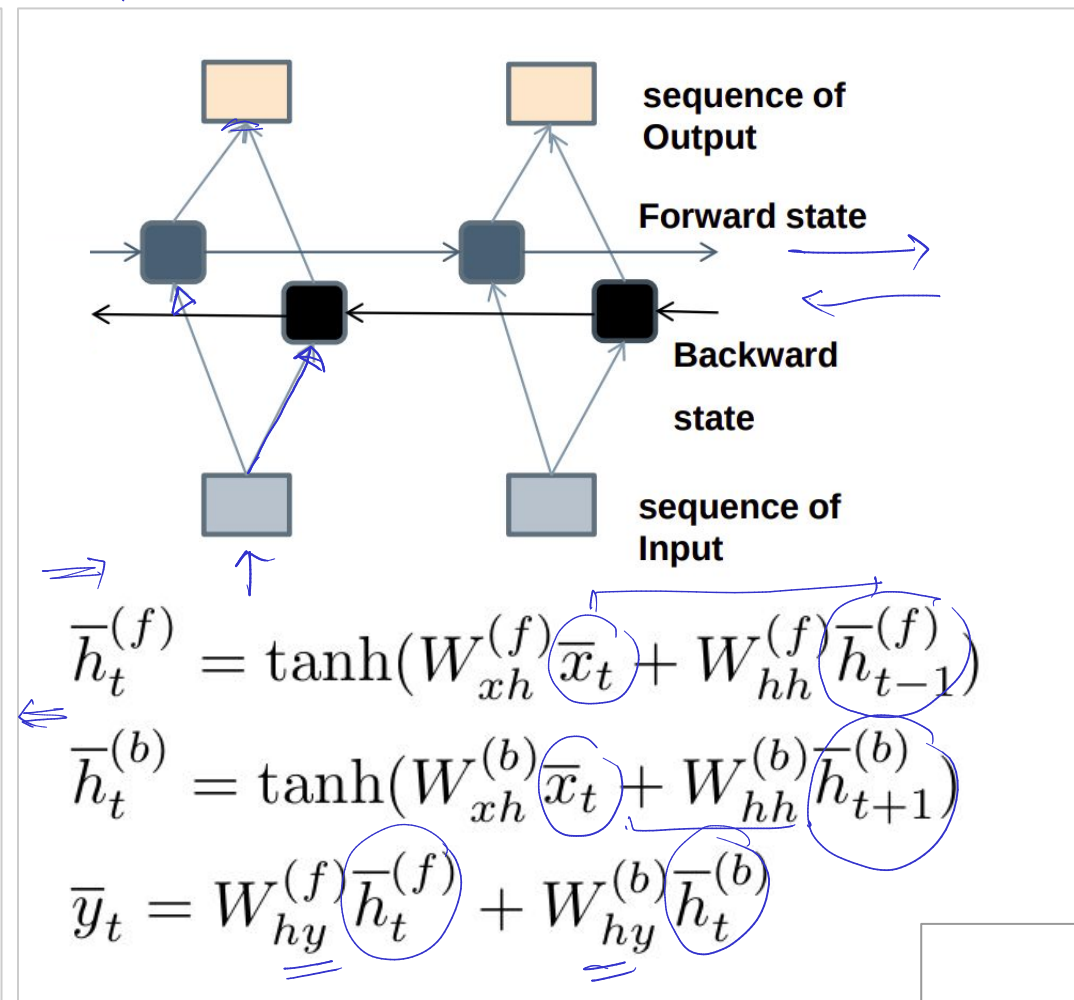
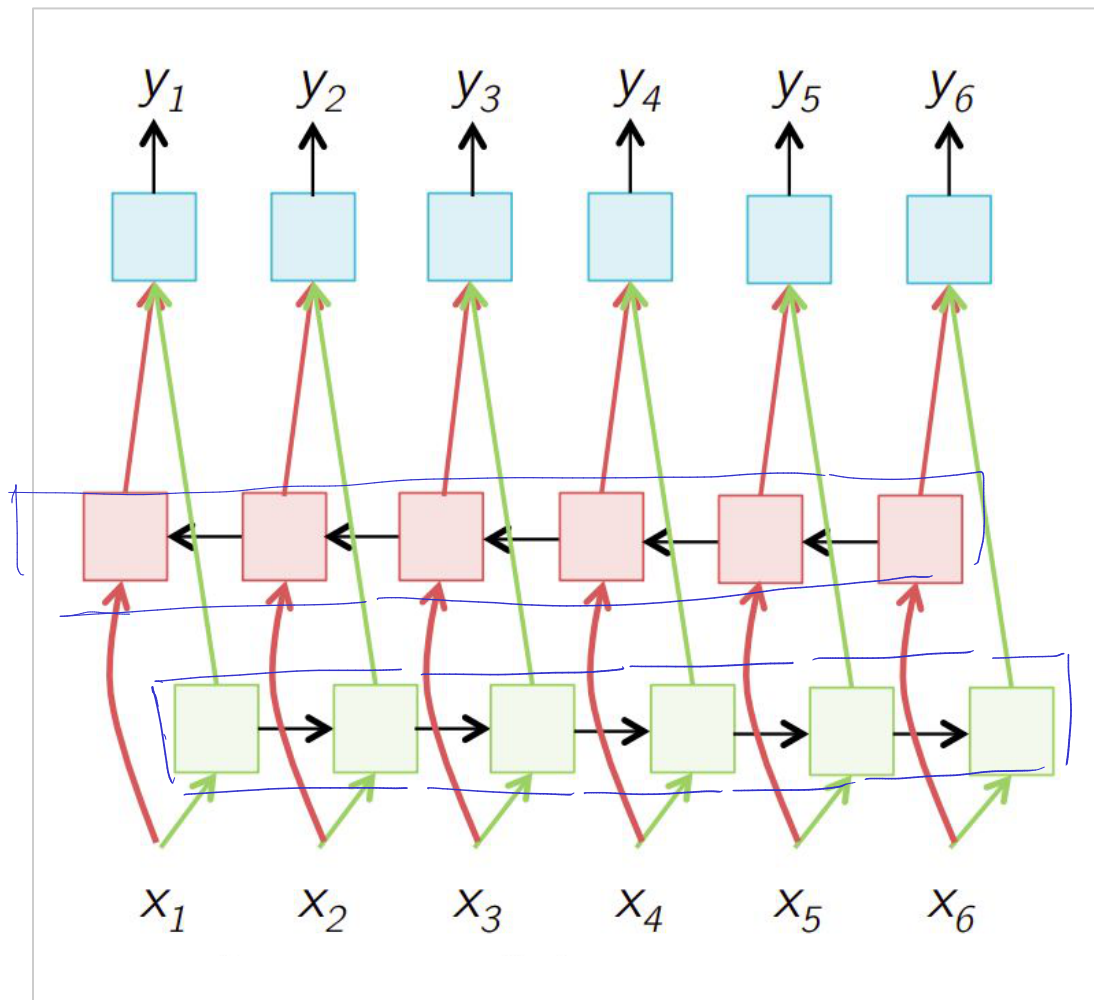
More distributed weights $\rightarrow$ RNN rep much more complex fns



Deep (stacked) recurrent neural network

A few simple variants of RNNs

very-very effective



Bidirectional recurrent neural network